

Vol 10 Chapter 5 — Fourier Analysis and Distributions

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 5 — Fourier Analysis and Distributions

Level 1 — one sentence

Fourier analysis decomposes functions into oscillatory modes (sums/integrals of $e^{i\omega t}$), Schwartz distributions extend differentiation and Fourier transformation to “objects like” $\delta(x)$ and $\delta'(x)$, and the substrate’s Wirtinger derivative (Vol 5 Ch 2) is the holomorphic-Fourier analog on D^5_V .

Level 2 — graduate-physicist precision

5.1 Fourier series

For periodic $f(x)$ with period L : $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n x / L}$ with $c_n = (1/L) \int_0^L f(x) e^{-2\pi i n x / L} dx$.

Parseval: $\int_0^L |f|^2 dx = L \sum |c_n|^2$.

Convergence: pointwise (Dirichlet), in L^2 (always for $f \in L^2$), uniformly (for smooth f).

5.2 Fourier transform

For $f \in L^1(\mathbb{R})$:

$$\hat{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int \hat{f}(k) e^{ikx} dk$$

Plancherel: $\int |f|^2 dx = (1/2\pi) \int |\hat{f}|^2 dk$ — Fourier transform is unitary on L^2 .

Properties: $\widehat{f'}(k) = ik\hat{f}(k)$ (differentiation $\leftrightarrow$ multiplication). $\widehat{\hat{f} \cdot g} = \hat{f} * \hat{g}/(2\pi)$ (multiplication $\leftrightarrow$ convolution).

Quantum mechanics: position and momentum representations are Fourier transforms of each other.

5.3 Distributions

Schwartz space $\mathcal{S}(\mathbb{R}^n)$ = rapidly decreasing smooth functions. Tempered distributions $\mathcal{S}'$ = continuous linear functionals on $\mathcal{S}$.

Examples: - $\delta(x)$: $\langle \delta, \phi \rangle = \phi(0)$ - $\delta'(x)$: $\langle \delta', \phi \rangle = -\phi'(0)$ (distributional derivative) - $\mathcal{P}(1/x)$: Cauchy principal value

Fourier transform extends to $\mathcal{S}'$: $\hat{\delta} = 1$, $\hat{1} = 2\pi\delta$, etc.

5.4 Substrate Wirtinger Fourier

The substrate's momentum operator is the Wirtinger derivative $-i\hbar\partial/\partial z_i$ on $H^2(D_{IV}^5)$ (Vol 5 Ch 2). This is the holomorphic-Fourier analog: position $\leftrightarrow$ momentum on the Bergman Hilbert space mirrors the standard Fourier transform on $L^2(\mathbb{R}^n)$.

5.5 K-audit anchors

- Vol 5 Ch 2: Wirtinger momentum on Bergman
- Reed and Simon Vol 1, 2: standard Fourier + distribution theory

Level 3 — 5th-grader accessibility

Fourier analysis: any function can be decomposed into sines/cosines (or $e^{i\omega t}$ exponentials). Convert between time domain and frequency domain. Used everywhere: signal processing, image compression (JPEG), quantum mechanics (position $\leftrightarrow$ momentum). Distributions extend Fourier transform to “delta function” $\delta(x)$ and derivatives. In BST: substrate momentum is the Wirtinger derivative — the holomorphic version of Fourier on the Bergman Hilbert space.

What comes next

Chapter 6 develops group theory and Lie groups.

Where to look this up

- Stein and Shakarchi, *Fourier Analysis*
- BST: Vol 5 Ch 2